

Maryem Ahmed

AI Engineer

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in LinkedIn 🐙 GitHub

Education

Bachelor of Computer Science and Information Systems

Aug 2022 – Jun 2026

Shorouk Academy

Experience

AI - Generative AI Professional

Nov. 2025 – Present

Digital Egypt Pioneers Initiative (DEPI)

- Developing expertise in Generative AI technologies including large language models (LLMs), prompt engineering, and AI-powered applications
- Building deep learning solutions with TensorFlow, Keras, and PyTorch, implementing CNNs, RNNs, and transformer architectures for various AI applications
- Applying MLOps practices for model deployment, monitoring, and lifecycle management in production environments
- Gaining hands-on experience in Natural Language Processing (NLP) and speech technologies, including text-to-speech and speech-to-text systems

AI - Data Science Trainee

Oct. 2024 – May 2025

Digital Egypt Pioneers - DEPI | Cairo

- Conducted exploratory data analysis (EDA) with Python to extract actionable insights, identify data patterns, and inform feature engineering decisions
- Built MLOps workflows incorporating version control, experiment tracking (MLflow), and model deployment best practices to ensure scalability and reproducibility
- Collaborated with cross-functional teams using Agile methodologies to deliver data-driven solutions aligned with business objectives

Projects

MeetGenie - AI Meeting Assistant | [GitHub](#) 

2025 – 2026

College Graduation Project · End-to-end LLM system

- Built an end-to-end AI pipeline that ingests raw meeting audio, transcribes it via WhisperX with speaker diarization, and stores speaker-turn chunks in ChromaDB for RAG-based question answering
- Designed a RAG pipeline using Groq's LLaMA 3.3 70B with dynamic top-k retrieval, a general-knowledge detector, and file-driven prompt templates editable without code changes
- Implemented structured NLP features including meeting summarisation, per-speaker personalised summaries, task extraction with inferred assignees, persistent task tracking, and natural language reminder saving
- Added full bilingual support (Egyptian Arabic + English) with automatic language detection; integrated ElevenLabs TTS with a custom Egyptian Arabic voice for spoken responses
- Benchmarked LLaMA 3.3 and Mixtral models against a mixed Arabic/English transcript dataset to evaluate summarisation and task extraction quality prior to production selection

NextHire - HR Assistant | [GitHub](#) 

2026

DEPI Graduation Project

- Built a fully offline RAG-based semantic search engine, fusing dense vector retrieval (ChromaDB + BGE-small embeddings) with BM25 sparse retrieval via Reciprocal Rank Fusion (RRF) for hybrid ranking
- Designed a section-aware PDF chunking pipeline (PyMuPDF + Tesseract OCR fallback) to extract structured sections — skills, experience, education — for higher-precision retrieval
- Optimized ingestion throughput with a persistent skip-list, enabling resumable processing of large CV datasets with zero redundant recomputation

Deep Learning Neural Network Project | [GitHub](#) 

- Designed and trained convolutional neural networks (CNNs) for image classification tasks, implementing

data augmentation techniques to improve model generalization

- Optimized neural network architectures through hyperparameter tuning and regularization methods, achieving high validation accuracy
- Developed comprehensive project documentation including model architecture diagrams, training procedures, and performance evaluation metrics

Sales Forecasting Using Time Series Models | [GitHub](#)

- Developed and optimized Light Gradient Boosting Machine and Prophet models for multi-store sales prediction, achieving RMSE reduction through hyperparameter tuning
- Implemented reproducible ML pipelines with automated data preprocessing, model training, and experiment tracking using MLflow, following MLOps best practices for model lifecycle management
- Engineered 15+ features from multi-source datasets (sales transactions, oil prices, holiday calendars) to capture seasonality, trend patterns, and external market influences

Technical Skills

Programming	Python, C, C++, C# Java
Generative AI & NLP	LLMs, RAG, Prompt Engineering, Semantic Search, Vector Databases, TTS/STT
Machine Learning & AI	Supervised/Unsupervised Learning, CNNs, Neural Networks, Feature Engineering
Databases	PostgreSQL, Microsoft SQL Server, MySQL, ChromaDB
ML Frameworks & Libraries	TensorFlow, Keras, PyTorch, scikit-learn, pandas, NumPy, Matplotlib
MLOps & Tools	MLflow, Docker, Git/GitHub, Model Deployment Pipelines

Certifications

- [DEPI - Data Scientist](#) 
- [Advanced Learning Algorithms](#) 
- [Supervised Machine Learning: Regression and Classification](#) 
- [Python Certificate – HackerRank](#) 
- [Machine Learning - NTI](#) 